

Robust Linear Regression via the Repeated Angular Shorth

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Abstract. Robust regression methods are designed to limit the effect of unusual observations, but they can still be misled when a large minority forms a coherent structure, such as a compact high-leverage cluster or a competing line. We propose *Repeated Angular Shorth Regression* (RAShR), a deterministic estimator based on pairwise chord directions rather than residuals. For each observation, the dominant local direction is obtained from the shortest angular interval containing half of its chord directions, and these local directions are aggregated in the same way. We establish exact-fit and equivariance properties and prove a separation result showing that sufficiently separated contamination does not affect the estimated direction. We also show that population LMS can be attracted toward a point-mass cluster near 50% contamination. Simulations indicate that RAShR is most useful under roughly 35%–46% coherent contamination, especially for compact leverage clusters, while MM-regression remains preferable in standard low-contamination settings.

Keywords: Robust regression · Pairwise slopes · Shorth · Leverage · Directional statistics · Breakdown.

1 Introduction

Fitting a straight line to data in the presence of unusual observations is a classical problem in statistics [6,14,10]. Such unusual observations are commonly called *outliers*. More generally, we use the term *contamination* to describe observations that do not follow the main pattern present in the data. Several regression methods have been developed to reduce the effect of such observations. Well-known examples include Least Median of Squares (LMS) [13], Least Trimmed Squares (LTS) [15], MM-estimation [20], and Siegel’s repeated median [17]. An important property of many robust estimators is their *breakdown point*. Informally, the breakdown point measures how large a fraction of the observations can be replaced by arbitrary values before the resulting estimate can be driven arbitrarily far away. For several classical robust regression estimators, this fraction can be close to one half.

A high breakdown point is therefore a strong form of protection against extreme contamination. However, it provides a worst-case guarantee about whether

the estimate remains bounded. It does not necessarily guarantee that the fitted line remains close to the linear pattern followed by most of the observations. This distinction becomes important when the unusual observations do not appear as isolated points, but instead form a structured group of their own. We refer to this situation as *coherent contamination*. Here, the contaminated observations are related to one another and may themselves form a compact cluster or follow a clear geometric pattern. An especially difficult case occurs when such a group lies at a *high-leverage* location. In simple linear regression, an observation has high leverage when its x -coordinate lies far from the main body of the data. Because such observations are far from the centre of the observed x -values, they can have a strong effect on the estimated slope.

Figure 1 illustrates this situation. The data contain 100 observations. Of these, 62 approximately follow the line

$$y = 1 + 2x,$$

while the remaining 38 observations form a compact cluster at a high-leverage location.

Although most of the observations follow a positive trend, OLS, LMS, and the MM-estimator all return lines with negative slopes in this example. For OLS, such sensitivity to high-leverage observations is expected. The behavior of LMS and MM-estimation may initially appear more surprising because these methods are specifically designed to tolerate substantial contamination. To understand what happens, recall that the *residual* of an observation is the difference between its observed y -value and the value predicted by a fitted line. Many regression procedures judge a candidate line, directly or indirectly, by the sizes of such residuals. We refer to methods whose fitting criterion is primarily based on these distances as *residual-based methods*.

A compact high-leverage cluster can create a misleading candidate line. A line passing close to the contaminating cluster may also pass sufficiently close to some observations from the majority group. As a result, it can achieve a favorable residual criterion even though most of the data were generated by a different line. In the case of LMS, for example, the criterion depends on a central order statistic of the squared residuals. A candidate line does not therefore need to fit the entire majority well; it only needs sufficiently many observations to have small residuals.

This example highlights the difference between two desirable properties. The first is preventing an estimate from becoming arbitrarily large under contamination. The second is identifying the line supported by the dominant structure in the data. A high breakdown point provides the first guarantee, but does not by itself ensure the second. The blue line in Figure 1 is produced by the method developed in this paper. At this stage, we show it only to illustrate the behavior that motivates our work. The estimator and its construction are introduced below and defined formally in Section 3. Situations of this kind are not purely artificial. A coherent group of unusual observations can arise when a batch of measurements is recorded using a different unit, when an instrument enters a saturated operating regime, when a physical process switches between two modes,

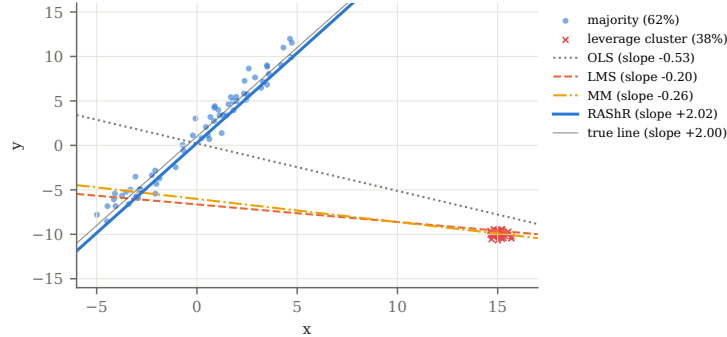


Fig. 1. A regression example with a compact high-leverage cluster containing 38% of the observations. Although 62% of the observations follow a positive-slope relationship, OLS, LMS, and the MM-estimator return negative slopes. The blue line is obtained using the method developed in this paper.

or when a subpopulation follows a different relationship from the rest of the data [14].

The example suggests that residual size alone may not always reveal which linear structure is supported by most of the data. We therefore study the problem from a different geometric point of view. Instead of asking only which line produces small residuals, we ask whether the directions determined by pairs of observations agree with one another.

To explain the idea, consider any two observations and the line segment joining them. We call this segment a *chord*. Every chord determines a direction, or equivalently an angle. If a large number of observations come from the same approximately linear population, then many chords formed between pairs of these observations have similar directions. Their angles therefore concentrate around the direction of the underlying line. The situation is different for chords that connect the majority observations to a compact contaminating cluster. These chords are viewed from many different locations in the majority group, and their directions can therefore be spread over a wider range of angles. Consequently, a contaminating cluster that is highly influential in residual space need not be equally dominant when the data are represented through pairwise directions.

This geometric observation motivates the estimator proposed in this paper, which we call *Repeated Angular Shorth Regression* (RAShR). Rather than selecting a line only according to residual size, RAShR looks for the direction receiving the strongest local agreement among pairwise chords.

The construction uses a simple statistical idea known as the *shorth*. Given a collection of values, a shorth is the shortest interval containing a specified fraction of them, traditionally one half. Such an interval identifies a region of high concentration without requiring the values outside that region to be discarded in advance. In RAShR, this idea is applied twice to angular data, producing a

two-stage or *repeated* angular shorth. Because a straight line does not have an intrinsic orientation, directions separated by 180° represent the same line. For example, an angle of 20° and an angle of 200° describe the same unoriented line. For this reason, the angular calculations are carried out on the *projective circle*, which treats these opposite directions as equivalent. The complete estimator, including its treatment of slope and intercept, is defined in Section 3. The ingredients used in this construction are related to several established ideas in statistics and geometric estimation. Theil–Sen regression [19,16] and Siegel’s repeated median [17] also use slopes obtained from pairs of observations. The shorth [1] and LMS [13] use shortest intervals or central subsets to obtain robustness. The Hough transform [5,3] and RANSAC [2] search for geometric structures that receive support from a substantial part of the data.

Our main contributions are as follows:

1. We propose a new regression estimator, called *Repeated Angular Shorth Regression* (RAShR), for situations in which a large and structured group of unusual observations can pull conventional regression procedures away from the trend followed by most of the data. Rather than relying only on residual size, RAShR uses agreement among pairwise chord directions to identify the dominant linear structure. Its two-stage angular shorth construction is designed to reduce the effect of a compact contaminating group while preserving the direction supported by the majority observations.
2. We provide a theoretical explanation for this behavior. In particular, we identify a geometric separation condition under which the chord directions associated with the contaminating group are separated in angle from those generated by the majority population. Under this condition, the contaminated observations make no contribution to the final RAShR direction estimate (Theorem 1). For comparison, we show that population LMS can be attracted toward a point-mass contaminating cluster as the contamination proportion approaches one half (Proposition 3).
3. We evaluate RAShR in a pre-declared simulation study against MM-regression, LTS, LMS, and a carefully tuned implementation of RANSAC. The experiments identify a range of approximately 35%–46% coherent contamination in which RAShR performs particularly well among the methods considered, with its clearest advantage occurring for compact high-leverage clusters. We also report the settings in which this advantage disappears or the method is not competitive.

The proposed estimator is intended for a specific robustness problem rather than as a general replacement for existing regression procedures. The purpose of RAShR is instead to address situations in which a large, geometrically coherent group of high-leverage observations can produce a misleading fit even when the estimator has a high breakdown point.

2 Related Work

Pairwise slopes have been used in robust regression for several decades. The slope estimator introduced by Theil [19] and later developed by Sen [16] is obtained by taking the median of all $\binom{n}{2}$ pairwise slopes. Its breakdown point is approximately 29%. A higher-breakdown alternative was introduced by Siegel [17] through the *repeated median*. For each observation, the median of the slopes joining that observation to all other observations is first computed. A second median is then taken over these observation-wise values:

$$\hat{\beta}_{\text{RM}} = \text{med}_i \text{med}_{j \neq i} \frac{y_j - y_i}{x_j - x_i}. \quad (1)$$

The repeated median achieves a breakdown point of 50%. Its asymptotic properties were subsequently studied in [4], and efficient algorithms for its computation have also been developed [11,12]. Another important line of work is based on selecting a subset of observations with small residuals. The Least Median of Squares (LMS) estimator was introduced by Rousseeuw [13]. It is obtained by minimizing the median of the squared residuals. Geometrically, in simple linear regression, this can be interpreted as finding a narrow strip that contains at least half of the observations [13,14]. The Least Trimmed Squares (LTS) estimator follows a related idea but minimizes the sum of a selected fraction of the smallest squared residuals. Practical algorithms for LTS are commonly based on concentration steps [15].

MM-estimation was introduced to combine a high breakdown point with improved statistical efficiency [20]. A high-breakdown initial estimate is first obtained, after which an M-estimation step is used to improve efficiency while retaining robustness. The theory and practical use of such estimators have been developed extensively in the robust-statistics literature; see, for example, [10]. Consensus-based approaches provide another way of fitting models in the presence of outliers. RANSAC was introduced by Fischler and Bolles [2]. Candidate models are generated from small randomly selected subsets of the data, and each candidate is evaluated according to the number of observations whose residuals fall below a chosen threshold. The model receiving sufficient consensus is then retained. Because of its simple geometric interpretation and ability to handle substantial numbers of outliers, RANSAC has been widely used in computer vision and related applications.

A related geometric approach is provided by the Hough transform [5]. Instead of fitting a line directly in the observation space, observations are mapped to a parameter space in which support for candidate lines can be accumulated. The statistical properties of Hough-transform-based estimators were studied by Goldenshluger and Zeevi [3]. Consistency and a high breakdown point were established, together with a cube-root convergence rate rather than the usual root- n rate. Mode-seeking estimators are also closely connected to the *shorth*. The shorth is based on the shortest interval containing a prescribed fraction, usually one half, of the observations. It was studied as a robust measure of location by Andrews et al. [1]. More generally, non-smooth mode-seeking procedures

of this type are known to exhibit cube-root asymptotic behavior under suitable conditions [7]. Regression has also been formulated using angular and directional information. Such formulations treat the orientation of a fitted line as an important geometric quantity rather than describing the fit solely through vertical residuals; examples and related developments are discussed in [8]. A different geometric interpretation is provided by support vector regression, in which an ε -insensitive tube is placed around the fitted regression function and deviations within this tube are not penalized [18].

3 Repeated Angular Shorth Regression

This section describes the proposed estimator in detail. The main idea is to represent the relationship between pairs of observations through their *directions* rather than through residuals. The procedure has three main steps. First, the two coordinate axes are placed on comparable scales. Second, a dominant direction is estimated separately from the point of view of each observation. Finally, these local directions are combined to obtain a single regression direction.

3.1 Standardization and projective chord directions

Let (x_i, y_i) , $i = 1, \dots, n$, denote the observed data points. Because the proposed method is based on angles, the relative scales of the x - and y -axes matter. For example, changing the unit of x from metres to centimetres changes the numerical angle of a line even though the underlying relationship between the variables has not changed. The two coordinates are therefore standardized before any angles are computed. A standardization used is:

$$u_i = \frac{x_i - \text{med}(x)}{s_x}, \quad v_i = \frac{y_i - \text{med}(y)}{s_y}, \quad (2)$$

where s_x and s_y are 1.4826 times the median absolute deviation (MAD) of the corresponding coordinates. The constant 1.4826 gives the usual normal-consistency scaling for the MAD. If a MAD is zero, the standard deviation of that coordinate is used as a fallback. After standardization, each pair of observations determines a line segment. As in the introduction, we refer to this segment as a *chord*. Its direction is described by the angle

$$\phi_{ij} = \text{atan2}(v_j - v_i, u_j - u_i) \bmod \pi. \quad (3)$$

The function atan2 computes the direction of the vector joining the two standardized observations. It is preferred to the ordinary slope ratio

$$\frac{v_j - v_i}{u_j - u_i}$$

because it remains well behaved when $u_j - u_i$ is very small, that is, when the chord is nearly vertical. There is one additional geometric point to consider. A

regression line has no preferred orientation: a direction ϕ and the opposite direction $\phi + \pi$ describe the same line. The chord directions are therefore represented on the *projective circle*, which may be identified with the interval $[0, \pi)$. The angular distance between two such directions a and b is defined as

$$d_\pi(a, b) = \min\{|a - b|, \pi - |a - b|\}. \quad (4)$$

This distance correctly treats angles close to 0 and π as being close to one another rather than far apart.

3.2 The angular shorth operator

The next step is to identify a region in angle space where many chord directions are concentrated. For this purpose, a circular version of the classical shorth is used. Let $A = \{a_1, \dots, a_m\} \subset [0, \pi)$ be a finite multiset of angles, and let $1 \leq h \leq m$. We define $\text{ASh}_h(A)$ as the midpoint of the shortest arc of the projective circle that contains at least h elements of A . The width of this shortest arc is denoted by $w_h(A)$. Unless stated otherwise, half of the available angles are used: $h = \lceil \frac{m}{2} \rceil$. For this default choice, we write simply $\text{ASh}(A)$. The idea is straightforward. Rather than taking the median angle, the operator searches for the narrowest angular interval containing at least half of the observations. Its midpoint is then used as the representative direction. In this way, the estimate is determined by a region of high local concentration.

The operator can be computed efficiently. First, the angles are sorted: $a_{(1)} \leq a_{(2)} \leq \dots \leq a_{(m)}$. Because the domain is circular, a second copy of the sorted angles shifted by π is appended. This converts intervals that cross the boundary at 0 into ordinary intervals on the real line. Among all windows containing h consecutive angles, the narrowest one is selected. If

$$r^* = \arg \min_r (a_{(r+h-1)} - a_{(r)}),$$

then

$$\text{ASh}_h(A) = \frac{1}{2} (a_{(r^*)} + a_{(r^*+h-1)}) \bmod \pi. \quad (5)$$

If more than one window has the same minimum width, the window with the smallest left endpoint is selected. This rule makes the estimator deterministic. The operator ASh can be viewed as a circular analogue of the classical shorth location estimator [1].

3.3 Local directions and repeated aggregation

The angular shorth is now applied in two stages. For each observation i , the point (u_i, v_i) is treated as an *anchor*. All chord directions joining this anchor to the remaining observations are collected: $\Phi_i = \{\phi_{ij} : j \neq i\}$. The local direction associated with anchor i is then defined by

$$\theta_i = \text{ASh}(\Phi_i). \quad (6)$$

At the same time, the width of the selected local angular interval is recorded:

$$w_i = w_{\lceil (n-1)/2 \rceil}(\Phi_i). \quad (7)$$

The value θ_i summarizes the dominant direction seen from observation i . A small value of w_i means that at least half of the chords from that anchor are concentrated in a narrow range of directions. A large value indicates weaker directional agreement. This first stage produces one local direction for every observation: $\theta_1, \dots, \theta_n$. A second angular shorth is then applied to these local directions:

$$\hat{\theta} = \text{ASh}\{\theta_1, \dots, \theta_n\}. \quad (8)$$

This second stage is the reason for the term *Repeated Angular Shorth Regression*: a shortest-half angular concentration is first found separately for each anchor, and the same type of operation is then applied again to the resulting local directions. The width of the outer angular shorth is also retained as a diagnostic. Together, the local widths w_i and the outer width indicate how strongly the data support a common direction.

3.4 Returning to the original scale

The direction $\hat{\theta}$ has been estimated in the standardized coordinate system. To obtain the regression slope on the original scale, $\hat{\theta}$ is first represented in the interval $(-\pi/2, \pi/2]$. The slope estimate is then

$$\hat{\beta} = \frac{s_y}{s_x} \tan \hat{\theta}. \quad (9)$$

Once the slope has been determined, the intercept is estimated robustly by taking the median of the individual intercepts implied by the observations:

$$\hat{\alpha} = \text{med}_i \left(y_i - \hat{\beta} x_i \right). \quad (10)$$

The final fitted line is therefore

$$\hat{y} = \hat{\alpha} + \hat{\beta}x.$$

Algorithm 1 summarizes the complete procedure.

Computational cost. For each of the n anchors, $n - 1$ chord directions are computed and sorted. Sorting requires $O(n \log n)$ time for each anchor, giving an overall time complexity of $O(n^2 \log n)$. If the anchors are processed one at a time, only $O(n)$ additional memory is required. Thus, the direct implementation has the same asymptotic time order as a direct implementation of Siegel's repeated median. In the implementations used in this study, it is also less expensive than an exhaustive search over elemental subsets for LMS or LTS, which requires $O(n^3)$ time.

Algorithm 1 Repeated Angular Shorth Regression**Require:** observations $(x_i, y_i)_{i=1}^n$, fraction $q = 0.5$

- 1: Robustly standardize the two coordinates using the median and MAD to obtain (u_i, v_i) ▷ Eq. (2)
- 2: **for** $i = 1, \dots, n$ **do**
- 3: Compute all chord directions

$$\Phi_i \leftarrow \{\text{atan2}(v_j - v_i, u_j - u_i) \bmod \pi : j \neq i\}$$

- 4: Sort Φ_i and append a second copy shifted by π
- 5: Find the narrowest angular window containing $\lceil q(n-1) \rceil$ directions
- 6: Store its midpoint as θ_i and its width as w_i
- 7: Apply the same shortest-window operation to $\{\theta_1, \dots, \theta_n\}$, using $\lceil qn \rceil$ values
- 8: Store the resulting midpoint as $\hat{\theta}$ and its width as w
- 9: Convert $\hat{\theta}$ back to the original coordinate scale:

$$\hat{\beta} = \frac{s_y}{s_x} \tan \hat{\theta}$$

- 10: Compute

$$\hat{\alpha} = \text{med}_i(y_i - \hat{\beta}x_i)$$

- 11: **return** $\hat{\alpha}, \hat{\beta}$, local widths $(w_i)_{i=1}^n$, and outer width w

4 Finite-Sample Properties

We next study some basic properties of RAShR. The purpose of this section is twofold. First, we verify that the estimator behaves correctly in simple cases: it recovers an exact straight line and responds in the expected way to changes of location and scale. Second, we describe a geometric condition under which a separated group of contaminated observations does not affect the estimated direction.

4.1 Exact fit and equivariance

We begin with the simplest possible setting, in which every observation lies exactly on the same straight line. A reasonable regression estimator should recover that line without error.

Proposition 1 (Exact fit). *Suppose*

$$y_i = \alpha^* + \beta^* x_i, \quad i = 1, \dots, n,$$

and suppose that the x_i 's are distinct. Then

$$(\hat{\alpha}, \hat{\beta}) = (\alpha^*, \beta^*).$$

Proof. After standardization, all observations still lie exactly on a straight line. Therefore every chord between two observations has the same projective direction,

$$\theta^* = \arctan\left(\frac{\beta^* s_x}{s_y}\right).$$

For every anchor, the shortest-half angular interval consequently has width zero and is centred at θ^* . The same is true at the outer aggregation stage, so $\hat{\theta} = \theta^*$. Using Eq. (9),

$$\hat{\beta} = \frac{s_y}{s_x} \tan \theta^* = \beta^*.$$

Finally, $y_i - \hat{\beta}x_i = \alpha^*$ for every observation. Their median is therefore α^* , giving $\hat{\alpha} = \alpha^*$.

Another desirable property is that the result should not depend on the units used to measure the two variables. For example, changing metres to centimetres should change the numerical value of the slope in the expected way, but should not otherwise change the fitted relationship.

Proposition 2 (Translation and positive-scale equivariance). *Let*

$$x'_i = a_x + b_x x_i, \quad y'_i = a_y + b_y y_i,$$

where $b_x, b_y > 0$. Then

$$\hat{\beta}' = \frac{b_y}{b_x} \hat{\beta}$$

and

$$\hat{\alpha}' = a_y + b_y \hat{\alpha} - \hat{\beta}' a_x.$$

The estimator is also invariant to a permutation of the observations.

Proof. The median is equivariant under translation and positive rescaling, while the MAD is equivariant under positive rescaling. Consequently, the standardized coordinates in Eq. (2) are unchanged by the transformation. The same pairwise directions are therefore obtained, and the same angular shorths are selected. Transforming the final direction back to the original units through Eq. (9) gives $\hat{\beta}' = \frac{b_y}{b_x} \hat{\beta}$. The stated intercept relation follows by substituting the transformed coordinates into the median-intercept formula. Permutation invariance follows because the angular shorth in Eq. (5) depends only on the multiset of angles after sorting, and not on the order in which the observations are supplied.

The estimator is not fully affine equivariant in the usual regression sense. In particular, replacing y by $y + \gamma x$ changes the marginal scale s_y , and this change does not in general correspond to a simple rotation of the standardized data. This limitation should be kept in mind when interpreting the estimator.

4.2 A separation theorem

The main robustness mechanism of RAShR comes from the shortest-half operation. If one group of angles forms a sufficiently tight cluster and the remaining angles are well separated from it, the shortest-half interval can select the concentrated group without being shifted toward the separated angles. We first state this property for a single angular shorth.

Lemma 1 (Separation). *Let A and B be finite multisets on the projective circle, and let $h \leq |A|$. Suppose that the shortest arc containing h elements of A is unique, with width $w_h(A)$. Assume further that every arc containing at least h elements of $A \cup B$, and containing at least one element of B , has width strictly larger than $w_h(A)$. Then*

$$\text{ASh}_h(A \cup B) = \text{ASh}_h(A).$$

Proof. Let I denote the unique shortest arc containing h elements of A . Its width is $w_h(A)$.

Suppose first that I contained an element of B . Then I would be an arc containing at least h elements of $A \cup B$, including at least one element from B , while still having width $w_h(A)$. This contradicts the assumption that every such arc must be strictly wider than $w_h(A)$. Therefore I contains no element of B . Hence I remains a feasible arc for $A \cup B$. Any other feasible arc that contains an element of B is strictly wider by assumption. A feasible arc containing only elements of A cannot be narrower than I , and by uniqueness it can have equal width only if it is I itself. Thus I is also the unique shortest feasible arc for $A \cup B$, proving the result.

The lemma gives a simple geometric message: adding a separated collection of angles does not change the angular shorth as long as every shortest feasible interval containing one of those additional angles is wider than the original shortest interval. We now apply this idea to the two stages of RAShR. Let G denote the set of observations belonging to the majority population, with $|G| > \frac{n}{2}$, and let $B = G^c$ denote the remaining observations. For a majority anchor $i \in G$, define the directions of chords joining it only to other majority observations by $\Phi_i^G = \{\phi_{ij} : j \in G \setminus \{i\}\}$. The two shortest-half sizes used by RAShR are

$$h_L = \left\lceil \frac{n-1}{2} \right\rceil, \quad h_O = \left\lceil \frac{n}{2} \right\rceil.$$

Because the majority contains more than half of the observations, $|G| - 1 \geq \left\lfloor \frac{n}{2} \right\rfloor \geq h_L$ and $|G| \geq h_O$. Thus the local and outer shorths can both be formed using majority observations alone. We consider the following two separation conditions.

(L) **Local separation.** For every $i \in G$, Lemma 1 applies with

$$A = \Phi_i^G, \quad B = \{\phi_{ij} : j \in B\}, \quad h = h_L.$$

In words, at every majority anchor, the shortest-half interval formed by majority-majority chords is narrower than any competing interval of the same size that involves a contaminated chord.

(O) **Outer separation.** Lemma 1 applies to the local directions with

$$A = \{\theta_i : i \in G\}, \quad B = \{\theta_i : i \in B\}, \quad h = h_O.$$

Thus, at the second stage, the concentrated majority-anchor directions form a narrower half-sample interval than any competing interval involving a contaminated-anchor direction.

Under these two conditions, the contaminated observations are excluded from both shortest-half selections.

Theorem 1 (Zero influence of separated contamination). *Suppose conditions (L) and (O) hold. Then*

$$\hat{\theta} = \text{ASh}_{h_O} \{ \text{ASh}_{h_L} (\Phi_i^G) : i \in G \}. \quad (11)$$

Thus, once the standardization scales are fixed, the estimated direction and hence $\hat{\beta}$ are determined entirely by the majority observations. Moving, adding, or removing contaminated observations does not change $\hat{\beta}$, provided that conditions (L) and (O) continue to hold, $|G| > n/2$, and the marginal medians and MADs used in the standardization remain unchanged.

Proof. For every majority anchor $i \in G$, condition (L) and Lemma 1 give $\theta_i = \text{ASh}_{h_L}(\Phi_i) = \text{ASh}_{h_L}(\Phi_i^G)$. Thus the local direction at every majority anchor depends only on chords joining majority observations. Condition (O) and Lemma 1 can then be applied a second time to the collection of local directions. This gives $\hat{\theta} = \text{ASh}_{h_O} \{ \theta_i : i \in G \}$, which is exactly Eq. (11).

The theorem describes a stronger property than merely keeping the estimate bounded. Under the stated separation conditions, the contaminated observations are not selected by either of the two angular shorth operations. Their locations therefore do not enter the final direction estimate, apart from the possible effect they may have on the initial marginal standardization.

This behavior is different from that of a median-based aggregation rule. Consider, for example, the inner median used by Siegel's repeated median. If all contaminated slopes lie on the same side of the majority slopes, they can change the rank position at which the median is taken. The resulting estimate can therefore change even when those contaminated slopes are far from the central group. The repeated median remains high-breakdown, but separation alone does not remove this rank effect.

4.3 Exact fit in the presence of contamination

The separation argument becomes especially simple when the majority observations lie exactly on one straight line.

Corollary 1 (Exact fit under contamination). *Suppose that the majority observations satisfy*

$$y_i = \alpha^* + \beta^* x_i, \quad i \in G,$$

with distinct x_i 's. Suppose also that no contaminated observation lies on this majority line and that

$$|B| < \left\lceil \frac{n-1}{2} \right\rceil.$$

Then

$$(\hat{\alpha}, \hat{\beta}) = (\alpha^*, \beta^*),$$

regardless of the positions of the contaminated observations.

Proof. At any majority anchor, every chord joining two majority observations has the same direction, say θ^* . Therefore these directions form a zero-width angular interval containing $|G| - 1$ elements. Since $|G| - 1 \geq h_L$, this interval is feasible for the local angular shorth. A chord joining a majority anchor to a contaminated point can have direction θ^* only if that contaminated point lies on the majority line, which has been excluded by assumption. Furthermore, any zero-width interval at another direction can contain at most $|B|$ contaminated chords. Because $|B| < h_L$, such an interval cannot satisfy the local half-sample requirement.

It follows that $\theta_i = \theta^*$ for every $i \in G$. At the outer stage, at least $|G| \geq h_O$ local directions are equal to θ^* . Since fewer than h_O directions arise from contaminated anchors, they cannot form a competing zero-width interval. Hence $\hat{\theta} = \theta^*$. Transforming back to the original scale gives $\hat{\beta} = \beta^*$. Finally, more than half of the residuals satisfy

$$y_i - \hat{\beta}x_i = \alpha^*,$$

so their median is α^* . Therefore $\hat{\alpha} = \alpha^*$.

This exact-fit property is not unique to RAShR. It is also shared by classical high-breakdown procedures such as LMS and the repeated median. The more interesting distinction arises when the majority observations contain noise, because the residual-based and directional criteria then respond differently to a coherent contaminating group.

4.4 Why a coherent cluster can attract a residual criterion

We now study this effect for population LMS. The purpose of the following result is not to describe every possible contamination pattern, but to isolate one mechanism through which a compact contaminating cluster can become attractive to a residual-based half-sample criterion. For a candidate line $\ell(x) = a + bx$, define

$$W(\ell) = \inf \left\{ w : \Pr(|Y - \ell(X)| \leq w) \geq \frac{1}{2} \right\}.$$

Thus $W(\ell)$ is the smallest half-width of a vertical strip around ℓ that contains at least half of the probability mass. Population LMS selects a line that minimizes this quantity. Consider the mixture model

$$(X, Y) \sim (1 - \varepsilon)P_G + \varepsilon\delta_c, \quad P_G : Y = \alpha^* + \beta^*X + e, \quad e \sim N(0, \sigma^2) \perp X, \quad (12)$$

where δ_c is a point mass at $c = (c_x, c_y)$. The majority population follows the line $\ell^*(x) = \alpha^* + \beta^*x$ with Gaussian vertical noise. The contaminating mass is placed at vertical distance $\Delta = |c_y - \alpha^* - \beta^*c_x| > 0$ from the majority line.

Proposition 3 (LMS is attracted by a point-mass cluster). *There exists $\varepsilon_0 < 1/2$ such that, for every $\varepsilon \in (\varepsilon_0, 1/2)$, every population LMS solution ℓ satisfies*

$$|c_y - \ell(c_x)| \leq W(\ell),$$

with

$$W(\ell) \longrightarrow 0 \quad \text{as } \varepsilon \uparrow \frac{1}{2}.$$

Consequently, the majority line ℓ^* is not an LMS solution for such contamination levels, and every LMS solution passes arbitrarily close to the contaminating point as ε approaches $1/2$.

Proof. Let w_ε be defined by $(1-\varepsilon) \Pr(|e| \leq w_\varepsilon) = \frac{1}{2}$. For $\varepsilon \geq 1/4$, $w_\varepsilon \geq w_{1/4} > 0$.

Now consider a strip around an arbitrary line ℓ that excludes the contaminating point c . Such a strip contains probability mass only from the good population. For the symmetric unimodal Gaussian error distribution, Anderson's inequality gives

$$\Pr(|e - t| \leq w) \leq \Pr(|e| \leq w)$$

for every shift t . Therefore any strip excluding c and containing half of the total probability mass must have half-width at least w_ε , and hence at least $w_{1/4}$ when $\varepsilon \geq 1/4$. Next, choose any non-vertical line ℓ_c passing through c . A strip of half-width w around ℓ_c already contains the entire point mass ε . Let

$$p(w) = P_G(|Y - \ell_c(X)| \leq w).$$

Because the Gaussian error has full support, $p(w) > 0$ for every $w > 0$. The strip contains at least half of the total probability mass whenever $(1-\varepsilon)p(w) \geq \frac{1}{2} - \varepsilon$. As $\varepsilon \uparrow 1/2$, the right-hand side approaches zero. Consequently, $W(\ell_c) \rightarrow 0$.

Choose $\varepsilon_0 \geq 1/4$ sufficiently large that $W(\ell_c) < \min\{w_{1/4}, \Delta\}$ for every $\varepsilon \in (\varepsilon_0, 1/2)$. An LMS solution ℓ must satisfy $W(\ell) \leq W(\ell_c) < w_{1/4}$. But every strip excluding c requires half-width at least $w_{1/4}$. Therefore the optimal strip must contain c , giving $|c_y - \ell(c_x)| \leq W(\ell)$. Since $W(\ell) \leq W(\ell_c) \rightarrow 0$, the LMS solution passes arbitrarily close to c as $\varepsilon \uparrow 1/2$. Finally, because $W(\ell_c) < \Delta$, the majority line ℓ^* , which is at distance Δ from c , cannot be an LMS solution in this regime.

The proposition gives a simple explanation for the behavior seen in Figure 1. As the contaminating point mass approaches one half of the population, a very narrow strip passing through that cluster needs to capture only a small additional amount of probability from the majority population in order to contain half of the total mass. The LMS criterion can therefore prefer a line close to the cluster even though the cluster remains a minority.

Related effects can also arise for other procedures that select or reward a subset of observations having small residuals. LTS and the high-breakdown initial scales used in MM procedures are relevant examples. However, the proposition above is proved specifically for population LMS, so no identical theoretical claim is made here for LTS or MM-estimation. Their behavior in the contamination settings studied in this paper is examined empirically in Section 5.

The proposition is asymptotic in the contamination fraction $\varepsilon \uparrow 1/2$. The finite-sample experiments later in the paper are used to determine how early this attraction mechanism becomes visible in practice.

Remark 1 (Why an “all chords” concentration condition is too strong). A stronger-looking theoretical condition could be obtained by assuming that every majority–majority chord lies within an angular distance $\delta < \pi/8$ of the true direction θ^* . Under such an assumption, one can derive a bound of the form $d_\pi(\hat{\theta}, \theta^*) \leq 4\delta$. Although mathematically valid, this assumption is not useful for the noisy regression settings considered here. When two observations have very similar x -coordinates, even a small amount of vertical noise can produce a chord with an almost arbitrary direction. Consequently, requiring *all* majority–majority chords to lie close to the true direction is generally far too restrictive.

5 Simulation Study

The simulation study was designed to determine the conditions under which RAShR provides an advantage over established robust regression methods. The comparison focuses on MM-regression, LTS, LMS, and RANSAC, with particular attention to large coherent groups of unusual observations.

Design. Unless stated otherwise, $n = 100$ observations are generated from $x \sim U[-5, 5]$, $y = 1 + 2x + e$, $e \sim N(0, 1)$. A fraction f of the observations is then replaced by one of three pre-declared contamination patterns. The first is a compact high-leverage cluster centred at $(15, -10)$ with Gaussian spread 0.3. The second is a competing line, $y = 5 - 1.25x + e_c$, $e_c \sim N(0, 0.5^2)$, over the same x -range. The third combines a heteroscedastic majority, with error standard deviation $0.2 + 0.6|x|$, and a competing line with noise standard deviation 0.3. The implementations and tuning constants are kept fixed across all scenarios. LTS is initialized from the ten best elemental fits and refined by 20 concentration steps. MM-regression uses the LTS fit as its starting point, followed by a MAD scale estimate and Tukey’s bisquare loss. RANSAC uses two-point samples, a MAD-based residual threshold, and 1000 trials. The large number of RANSAC trials was chosen deliberately so that the comparison would not be weakened by insufficient random sampling. The primary accuracy measure is the absolute slope error $|\hat{\beta} - 2|$. For the coherent-contamination experiments, we also report the failure rate $\Pr(|\hat{\beta} - 2| > 0.5)$. This second measure is useful because, in the difficult contamination regime, an estimator often either remains close to the majority line or moves toward the competing structure.

Standard settings. We first examined clean data and more conventional contamination patterns. These experiments show that RAShR is not uniformly preferable to existing robust methods. On clean data, its median absolute slope error is 0.033, compared with 0.024 for MM-regression and 0.022 for RANSAC. Under 20% vertical contamination, the corresponding errors are 0.040, 0.027, and 0.037. With 10% bad leverage, MM and RANSAC again perform slightly better, with errors 0.020 and 0.021, compared with 0.041 for RAShR. A similar pattern is seen for a 20% competing line and for Cauchy errors. These results

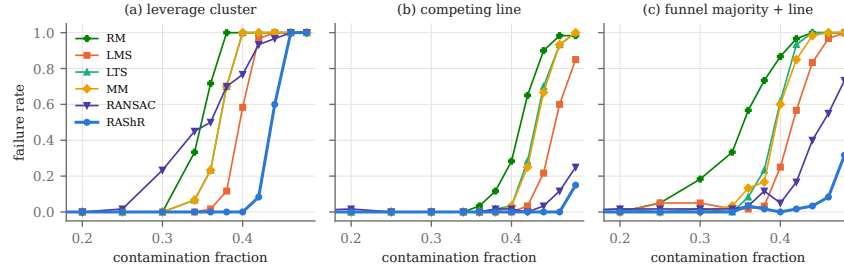


Fig. 2. Failure rate $\Pr(|\hat{\beta} - 2| > 0.5)$ as the coherent contamination fraction increases. Each point is based on 60 replications. The three panels show a compact high-leverage cluster, a competing line, and a heteroscedastic majority with a competing line.

establish an important limitation: at low contamination levels or in standard robust-regression settings, MM-regression is generally the more efficient choice. The advantage of RAShR appears only when the contaminating group becomes both large and geometrically coherent. The full standard-scenario results support this limitation.

Near-half coherent contamination. Figure 2 shows what happens as the fraction of coherent contamination increases. Below approximately 30%, the high-breakdown methods behave similarly. The methods begin to separate clearly between about 35% and 46%.

The strongest difference occurs for the compact high-leverage cluster. At 36% contamination, RAShR has no failures, while the failure rates are 2% for LMS, 23% for LTS and MM, and 50% for RANSAC. At 40%, LTS and MM fail in every replication, LMS fails in 58%, and RANSAC in 77%, while no RAShR failure is observed. At 44%, RAShR still succeeds in 40% of the samples, whereas every competing method fails in at least 97%.

The competing-line experiment shows a similar, although less extreme, pattern. RAShR has no failures through 46% contamination. At 46%, the failure rates are 60% for LMS, 93% for LTS and MM, and 12% for RANSAC. Thus RANSAC remains a strong competitor for this geometry, but its failure rate also begins to increase near the half-contamination boundary.

The heteroscedastic experiment gives the same qualitative conclusion. At 40% contamination, the failure rates are 25% for LMS, 62% for LTS, 60% for MM, and 5% for RANSAC, while no RAShR failure is observed. At 44%, the corresponding rates become 83%, 100%, 98%, 40%, and 3%, respectively. Even at 46%, RAShR fails in only 8% of the replications.

The advantage is therefore not a general “near-50%” guarantee. For the compact leverage cluster, all methods fail at 46%. Rather, the experiments identify a finite regime, approximately 35%–46%, in which RAShR can remain close to the majority line after several established methods have begun to follow the compet-

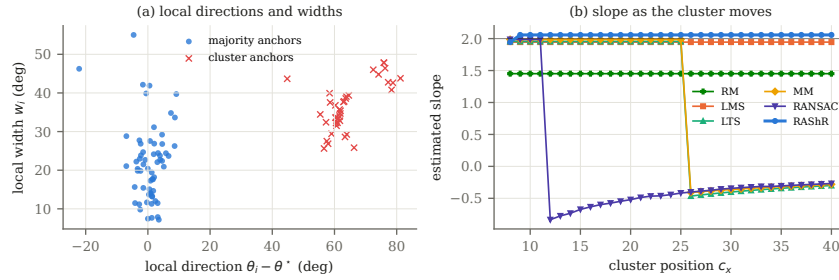


Fig. 3. Geometric behavior of RAShR. Left: local directions and widths for the example of Figure 1; majority anchors form the larger concentrated group. Right: estimated slope as a fixed 36-point cluster is moved horizontally while the majority sample is held fixed.

ing structure. These numerical patterns are those reported in the contamination sweeps.

When does the advantage disappear? The compact-cluster result was tested further by changing the cluster location, spread, and sample size. A 38% cluster was placed in twelve different directions and at two distances from the majority centre. In ten of the twelve directions, RAShR had no failures at either distance. The two difficult directions were approximately 30° and -150° , where the cluster lies close to the extension of the majority line.

This failure has a simple geometric explanation. When the cluster is well separated from the majority trend, chords joining majority observations to the cluster occupy different directions from the main group of majority-majority chords. When the cluster lies almost along the extension of the majority line, these two sets of directions begin to overlap and the angular separation is lost. At distance 30, the failure rate of RAShR increases to 78% in the 30° direction and 75% in the -150° direction. The competing methods also perform poorly in these directions. Compactness is equally important. For a very tight cluster with spread 0.05, no RAShR failures are observed, while LMS fails in 25% and LTS and MM in about 98% of the samples. When the spread is increased to 1 or 3, however, the cluster no longer behaves as a compact coherent group. In this case LMS, LTS, MM, and RAShR all fail in at most 2% of the samples. The advantage of RAShR is therefore specific to compact coherent contamination rather than arbitrary outliers. The same compact-cluster behavior is observed for all tested sample sizes $n \in \{40, 60, 200, 400\}$. Figure 3 also illustrates why this happens. Majority anchors produce a large group of local directions close to the true slope, whereas cluster anchors form a smaller group at a different direction. The outer angular shorth therefore selects the majority group. In the horizontal cluster-movement experiment, the RAShR slope becomes exactly constant at 2.058 once the cluster is sufficiently separated, which is consistent with the mechanism described by Theorem 1.

RAShR is not the preferred method for clean data or low contamination, where MM-regression is generally more efficient. Its advantage appears when roughly one-third to nearly one-half of the observations form a coherent competing structure. The clearest improvement occurs for compact high-leverage clusters; the advantage becomes weaker when the cluster is diffuse or nearly aligned with the majority line. These results directly support the performance claim stated in the Introduction.

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References

1. Andrews, D.F., Bickel, P.J., Hampel, F.R., Huber, P.J., Rogers, W.H., Tukey, J.W.: Robust Estimates of Location: Survey and Advances. Princeton University Press, Princeton (1972)
2. Fischler, M.A., Bolles, R.C.: Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Commun. ACM* **24**(6), 381–395 (1981). <https://doi.org/10.1145/358669.358692>
3. Goldenshluger, A., Zeevi, A.: The Hough transform estimator. *Ann. Stat.* **32**(5), 1908–1932 (2004). <https://doi.org/10.1214/009053604000000760>
4. Hössjer, O., Rousseeuw, P.J., Croux, C.: Asymptotics of the repeated median slope estimator. *Ann. Stat.* **22**(3), 1478–1501 (1994). <https://doi.org/10.1214/aos/1176325638>
5. Hough, P.V.C.: Method and means for recognizing complex patterns. US Patent 3,069,654 (1962)
6. Huber, P.J.: Robust estimation of a location parameter. *Ann. Math. Stat.* **35**(1), 73–101 (1964). <https://doi.org/10.1214/aoms/1177703732>
7. Kim, J., Pollard, D.: Cube root asymptotics. *Ann. Stat.* **18**(1), 191–219 (1990). <https://doi.org/10.1214/aos/1176347498>
8. Komsta, L.: Directional median analysis. *Chemom. Intell. Lab. Syst.* **153**, 110–115 (2016). <https://doi.org/10.1016/j.chemolab.2016.03.005>
9. Maechler, M., Rousseeuw, P., Croux, C., Todorov, V., Ruckstuhl, A., Salibián-Barrera, M., Verbeke, T., Koller, M., Conceicao, E.L.T., di Palma, M.A.: robustbase: Basic Robust Statistics. R package. <https://doi.org/10.32614/CRAN.package.robustbase> <https://CRAN.R-project.org/package=robustbase>
10. Maronna, R.A., Martin, R.D., Yohai, V.J., Salibián-Barrera, M.: Robust Statistics: Theory and Methods (with R), 2nd edn. Wiley (2019). <https://doi.org/10.1002/9781119214656>
11. Matoušek, J., Mount, D.M., Netanyahu, N.S.: Efficient randomized algorithms for the repeated median line estimator. *Algorithmica* **20**(2), 136–150 (1998). <https://doi.org/10.1007/PL00009190>
12. Raymaekers, J.: robslopes: Efficient Computation of the (Repeated) Median Slope. *R J.* **14**(4), 38–49 (2022). <https://doi.org/10.32614/RJ-2023-012>

13. Rousseeuw, P.J.: Least median of squares regression. *J. Am. Stat. Assoc.* **79**(388), 871–880 (1984). <https://doi.org/10.1080/01621459.1984.10477105>
14. Rousseeuw, P.J., Leroy, A.M.: *Robust Regression and Outlier Detection*. Wiley, New York (1987). <https://doi.org/10.1002/0471725382>
15. Rousseeuw, P.J., Van Driessen, K.: Computing LTS regression for large data sets. *Data Min. Knowl. Discov.* **12**(1), 29–45 (2006). <https://doi.org/10.1007/s10618-005-0024-4>
16. Sen, P.K.: Estimates of the regression coefficient based on Kendall’s tau. *J. Am. Stat. Assoc.* **63**(324), 1379–1389 (1968). <https://doi.org/10.1080/01621459.1968.10480934>
17. Siegel, A.F.: Robust regression using repeated medians. *Biometrika* **69**(1), 242–244 (1982). <https://doi.org/10.1093/biomet/69.1.242>
18. Smola, A.J., Schölkopf, B.: A tutorial on support vector regression. *Stat. Comput.* **14**(3), 199–222 (2004). <https://doi.org/10.1023/B:STCO.0000035301.49549.88>
19. Theil, H.: A rank-invariant method of linear and polynomial regression analysis. I. *Indag. Math.* **12**, 85–91 (1950)
20. Yohai, V.J.: High breakdown-point and high efficiency robust estimates for regression. *Ann. Stat.* **15**(2), 642–656 (1987). <https://doi.org/10.1214/aos/1176350366>